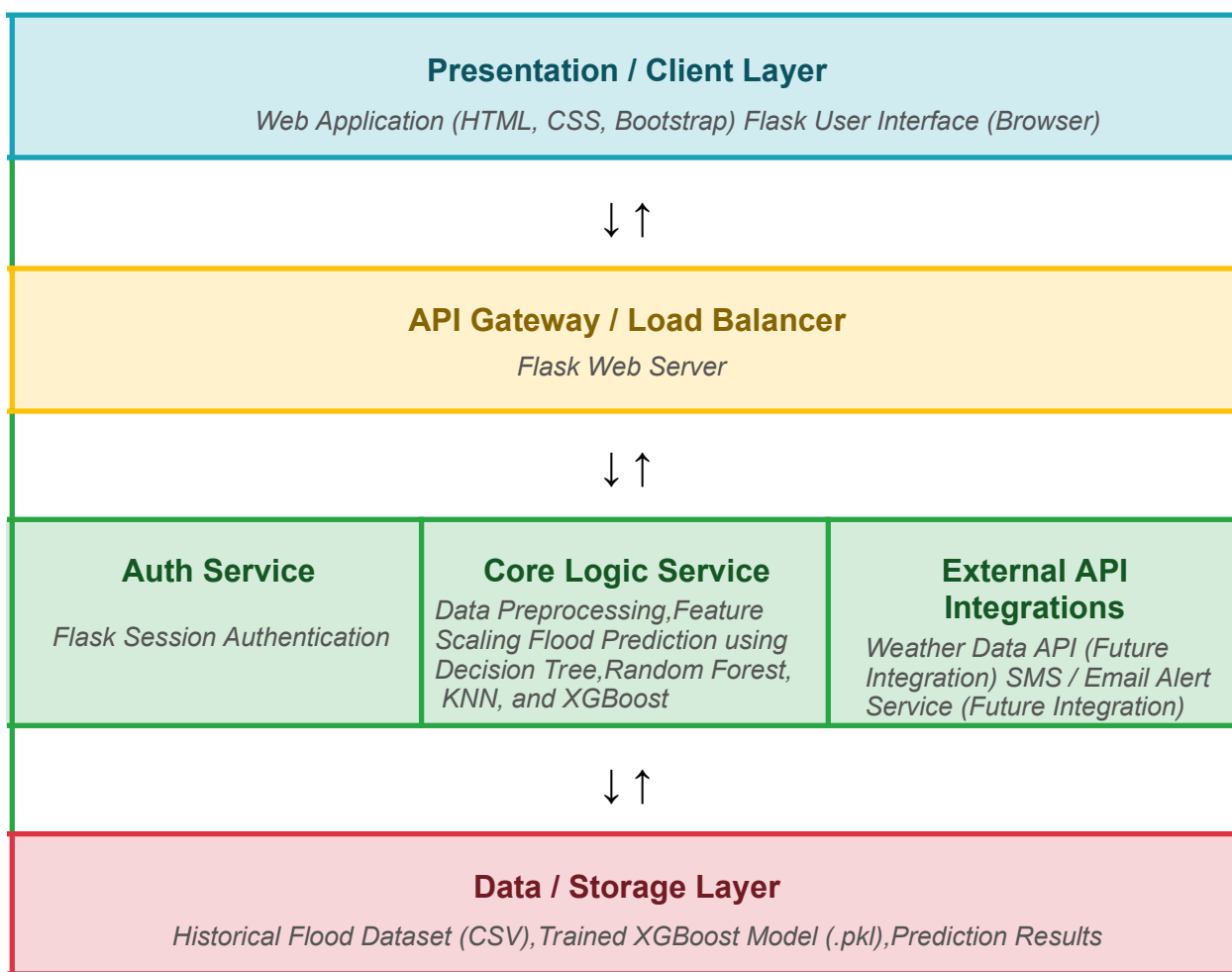


Solution Architecture

Date	27 June 2026
Team ID	
Project Name	Raising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface for entering rainfall and weather parameters and displaying flood prediction results.	HTML, CSS, Bootstrap
Flask Web Server	Receives user inputs, sends them to the ML model, and returns prediction results to the interface.	Flask, Python
Machine Learning Prediction Engine	Processes input data, performs preprocessing, and predicts flood occurrence using trained classification models.	Scikit-learn, XGBoost
Data Storage	Stores historical weather dataset and trained machine learning model.	CSV Dataset, Pickle (.pkl)